

15.6 Viscosity and Turbulence

LO 15.6 *Qualitatively describe the roles of viscosity and turbulence.*

Moving fluid interacts with the surfaces it contacts, resulting in a kind of fluid friction called **viscosity**. Viscosity also results from the transfer of momentum among adjacent layers within a fluid. Viscosity is especially important right near fluid boundaries because viscous forces bring the fluid to a complete stop at the boundary (Fig. 15.20). This boundary effect produces drag forces on objects moving through fluids—but it's the same drag at the surfaces of airplane and ship propellers that exerts a force on the fluid. Without viscosity, propellers would spin uselessly and planes and ships would go nowhere.

Viscosity depends on fluid properties and dimensions. Honey is more viscous than water, but at the tiny scales of a human capillary or a bacterium wiggling its flagella for propulsion, water too can be extremely viscous. Viscosity is also important in stabilizing flows that would otherwise become **turbulent**, or chaotically unsteady. Turbulence results from the growth of waves that gain energy at the expense of the flow, turning a smooth flow into a chaotic mess (Fig. 15.21). Turbulence is still not fully understood and presents ongoing challenges to scientists and engineers.

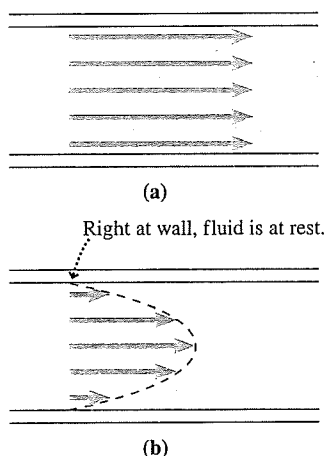


FIGURE 15.20 Velocity profiles in flows that are (a) inviscid (without viscosity) and (b) viscous.



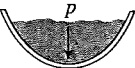
FIGURE 15.21 Smooth flow becomes turbulent, shown here in a column of rising smoke.

Chapter 15 Summary

Big Idea

Fluid is matter that readily deforms and flows under the influence of forces. Pressure, density, and flow velocity characterize fluids. Liquids and slowly moving gases are **incompressible**, meaning their density is essentially constant. A fluid that isn't moving is in **hydrostatic equilibrium**. In the presence of gravity, equilibrium requires that fluid pressure increase with depth.


A solid
maintains its shape.

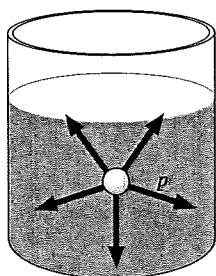

Increasing
A liquid takes
the shape of its container.


A gas fills
a closed container.

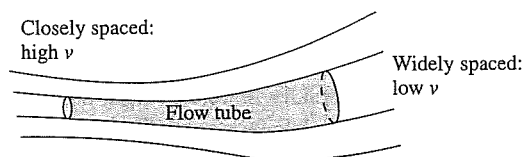
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Key Concepts and Equations

Pressure is the force per unit area: $p = F/A$. The pressure in a fluid exerts itself equally in all directions.



Streamlines represent a moving fluid.



The **continuity equation** describes the conservation of mass along a flow tube:

$$\rho v A = \text{constant (any fluid)}$$

$$v A = \text{constant (incompressible fluid)}$$

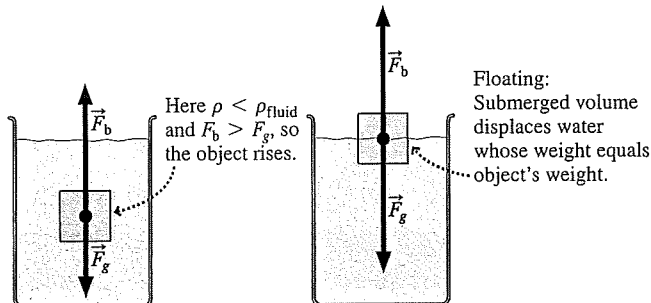
Bernoulli's equation describes the conservation of energy:

$$p + \frac{1}{2}\rho v^2 + \rho g y = \text{constant (incompressible fluid, neglecting viscosity)}$$

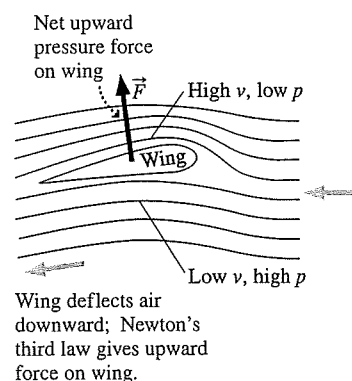
Viscosity, or fluid friction, is especially important when fluids interact with solid objects.

Applications

Archimedes' principle states that the **buoyancy force** $\vec{F}_b$ due to pressure on an object has the same magnitude as the weight of the displaced fluid. For an object less dense than a fluid, the buoyancy force exceeds gravity and the object floats; otherwise, it sinks or is in neutral buoyancy.



Bernoulli's principle helps explain lift forces, although ultimately these are based in Newton's third law.



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BIO Biology and/or medicine-related problems **DATA** Data problems **ENV** Environmental problems **CH** Challenge problems **COMP** Computer problems



Learning Outcomes After finishing this chapter you should be able to:

- | | |
|---|---|
| <p>LO 15.1 Characterize fluids by density and pressure.
For Thought and Discussion Questions 15.1, 15.2; Exercises 15.11, 15.12, 15.13, 15.14, 15.15, 15.16, 15.21; Problems 15.40, 15.41, 15.42, 15.67, 15.77</p> <p>LO 15.2 Describe how fluid pressure varies in hydrostatic equilibrium.
For Thought and Discussion Questions 15.3, 15.6, 15.7; Exercises 15.17, 15.18, 15.19, 15.20, 15.22; Problems 15.43, 15.44, 15.45, 15.46, 15.47, 15.63, 15.64, 15.65, 15.72, 15.73, 15.74, 15.75, 15.76, 15.78</p> <p>LO 15.3 Use Archimedes' principle as it applies to floating and submerged objects.</p> | <p>For Thought and Discussion Questions 15.4, 15.5, 15.8, 15.9; Exercises 15.23, 15.24, 15.25, 15.26; Problems 15.48, 15.49, 15.50, 15.51, 15.52, 15.53, 15.54, 15.60, 15.70, 15.79</p> <p>LO 15.4 Apply conservation of mass and energy in fluid dynamics.
Exercises 15.27, 15.28, 15.29, 15.30, 15.31; Problem 15.66</p> <p>LO 15.5 Apply Bernoulli's principle to applications in fluid dynamics.
For Thought and Discussion Question 15.10; Problems 15.55, 15.56, 15.57, 15.58, 15.59, 15.61, 15.62, 15.68, 15.69, 15.71</p> <p>LO 15.6 Qualitatively describe the roles of viscosity and turbulence.</p> |
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For Thought and Discussion

1. Why do your ears “pop” when you drive up a mountain?
2. You have two spheres of the same volume, one made of steel and the other of wood. Which one do you expect to float in a pool of water? Why?
3. The three containers in Fig. 15.22 are filled to the same level and are open to the atmosphere. How do the pressures at the bottoms of the three containers compare?

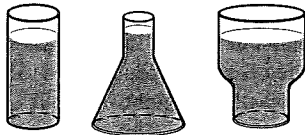


FIGURE 15.22 For Thought and Discussion 3.

4. Why is it easier to float in the ocean than in fresh water?
5. Figure 15.23 shows a cork suspended from the bottom of a sealed container of water. The container is on a turntable rotating about a vertical axis, as shown. Explain the position of the cork.

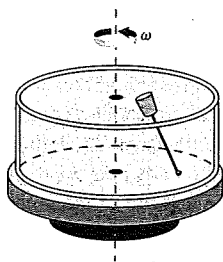


FIGURE 15.23 For Thought and Discussion 5.

6. Why are dams thicker at the bottom than at the top?
7. Mercury is used in barometers to measure atmospheric pressure. Why is Mercury a better option than water?
8. A helium-filled balloon stops rising long before it reaches the “top” of the atmosphere, but a cork released from the bottom of a lake rises all the way to the surface. Why the difference?
9. A barge filled with steel beams overturns in a lake, spilling its cargo. Does the water level in the lake rise, fall, or remain the same?
10. Why do airplanes take off into the wind?

Exercises and Problems

Exercises

Section 15.1 Density and Pressure

11. The density of molasses is 1600 kg/m^3 . Find the mass of the molasses in a 0.95-L jar.
12. Atomic nuclei have densities around 10^{17} kg/m^3 , while water’s density is 10^3 kg/m^3 . Roughly what fraction of water’s volume is *not* empty space?
13. Compressed air with mass 9.8 kg is stored in a 0.041-m^3 cylinder. (a) What’s the density of the compressed air? (b) What volume would the same gas occupy at a typical atmospheric density of 1.2 kg/m^3 ?
14. What’s the weight of a column of air with cross-sectional area 1 m^2 extending from Earth’s surface to the top of the atmosphere?

15. The *diamond anvil* is used by scientists and engineers to study matter under extreme pressures, simulating conditions such as those found at the centers of planets. A typical anvil consists of two diamonds with parallel faces measuring some $200 \mu\text{m}$ in diameter. The sample under study is placed between the diamonds, and a force is applied to the diamonds. Estimate the pressure that results when the force on the diamonds is 6 kN.
16. You unbend a paper clip made from 0.8-mm-diameter wire and push the end against the wall. What force must you apply to give a pressure of 135 atm?

Section 15.2 Hydrostatic Equilibrium

17. What’s the density of a fluid whose pressure increases at the rate of 100 kPa for every 8.0 m of depth?
18. A research submarine can withstand an external pressure of 57.5 MPa when its internal pressure is 101.3 kPa. How deep can it dive?
19. Scuba equipment provides a diver with air at the same pressure as the surrounding water. But at pressures higher than about 1 MPa, the nitrogen in air becomes dangerously narcotic. At what depth does nitrogen narcosis become a hazard?
20. A vertical tube open at the top contains 5.0 cm of oil with density 0.82 g/cm^3 , floating on 5.0 cm of water. Find the gauge pressure at the bottom of the tube.
21. A child attempts to drink water through a 36-cm-long straw but finds that the water rises only 25 cm. By how much has the child reduced the pressure in her mouth below atmospheric pressure?
22. Barometric pressure in the eye of a hurricane is 0.94 atm (714.4 mm of mercury). How does the level of the ocean surface under the eye compare with the level under a distant fair-weather region where the pressure is 1.0 atm?

Section 15.3 Archimedes’ Principle and Buoyancy

23. On land, the most massive concrete block you can carry is 25 kg. Given concrete’s 2200-kg/m^3 density, how massive a block could you carry underwater?
24. An 8.3-g jewel has apparent weight 56 mN when submerged in water. Could the jewel be a sapphire (density 3.98 g/cm^3)?
25. Styrofoam’s density is 160 kg/m^3 . What percent error is introduced by weighing a Styrofoam block in air (density 1.2 kg/m^3), which exerts an upward buoyancy force, rather than in vacuum?
26. A steel drum has volume 0.23 m^3 and mass 16 kg. Will it float in water when filled with (a) water or (b) gasoline (density 860 kg/m^3)?

Sections 15.4 and 15.5 Fluid Dynamics and Applications

27. Water flows through a 2.6-cm-diameter pipe at 1.7 m/s. If the pipe narrows to 1.6-cm diameter, what’s the flow speed in the constriction?
28. Show that pressure has the units of energy density.
29. A typical mass flow rate for the Nile River is $2.8 \times 10^7 \text{ kg/s}$. Find (a) the volume flow rate and (b) the flow speed in a region where the river is 1.8 km wide and averages 8.2 m deep.
30. A fire hose 10 cm in diameter delivers water at 22 kg/s. The hose terminates in a 2.1-cm-diameter nozzle. What are the flow speeds (a) in the hose and (b) at the nozzle?
31. A typical human aorta, the main artery from the heart, is 1.8 cm in diameter and carries blood at 35 cm/s. Find the flow speed around a clot that reduces the flow area by 80%.

Example Variations

The following problems are based on two worked examples from the text. Each set of four problems is designed to help you make connections that enhance your understanding of physics and to build your confidence in solving problems that differ from ones you’ve seen before.

The first problem in each set is essentially the example problem but with different numbers. The second problem presents the same scenario as the example but asks a different question. The third and fourth problems repeat this pattern but with entirely different scenarios.

32. **Example 15.4** An iceberg derived from a Greenland glacier contains gravel entrained in the ice as it moved across the land before calving into the sea. The entrained gravel increases the iceberg's density to 952 kg/m^3 . What fraction of the iceberg's volume is submerged when it's floating in ocean water with density 1030 kg/m^3 ?
33. **Example 15.4** An iceberg has mass 138,000 tonnes (1 tonne = 1000 kg) and is composed of pure ice with density 917 kg/m^3 and entrained gravel from rock with density 2750 kg/m^3 . If 95.5% of the iceberg's volume is submerged, how much of its mass is ice and how much is rock? Take the density of seawater to be 1030 kg/m^3 .
34. **Example 15.4** The Michelson–Morley experiment (see Chapter 33) was a very precise optical experiment that helped pave the way for Einstein's theory of relativity. To isolate it from external vibrations, the entire experiment was mounted on a square slab of sandstone 0.30 m thick and 1.5 m on a side, with a mass of 1.7 tonnes (1700 kg). The slab, in turn, floated in a trough of liquid mercury (density 13.69 tonnes/m^3). What percentage of the slab's volume was below the surface of the mercury?
35. **Example 15.4** Lakes on Saturn's moon Titan are filled with liquid hydrocarbons—largely methane but also some ethane. Suppose scientists want to send a probe to explore Titan's largest lake (which is about the size of Earth's Lake Ontario). The probe is to be a cylinder 56.3 cm in diameter, with a mass of 135 kg, and it's supposed to float with the long dimension vertical, half submerged and half above the lake surface. If the liquid in the lake has density 482 kg/m^3 , what should be the probe's length?
36. **Example 15.6** A large cylindrical tank is filled with water to a depth of 2.68 m. There's a small-diameter pipe emerging from the bottom of the tank, with a valve. If you open the valve, at what speed will water flow from the tank?
37. **Example 15.6** You'd like to determine the depth of water in a large tank, but you don't have any way to measure it directly. However, there's a small-diameter pipe emerging from the bottom of the tank, and when you open the valve on this pipe, you observe water spraying out at 5.46 m/s. What's the depth of the water in the tank?
38. **Example 15.6** A sealed tank holds water to a depth of 2.68 m. Above the water is air, pressurized to 186 kPa. If you open a small hole in the bottom of the tank, exposing the water at the bottom to standard atmospheric pressure, at what speed will the water initially emerge?
39. **Example 15.6** A fire extinguisher consists of a water-filled sealed canister with a hose whose diameter is much smaller than that of the canister. What should be the pressure of the water if it's to emerge initially from the hose at 18.8 m/s? Assume normal atmospheric pressure at the hose nozzle, and neglect pressure variation with height inside the container.

Problems

40. When a couple with total mass 150 kg lies on a water bed, pressure in the bed increases by 4700 Pa. What surface area of the two bodies is in contact with the bed?
41. A fully loaded Volvo station wagon has mass 1950 kg. If each of its four tires is inflated to a gauge pressure of 270 kPa, what's the total tire area in contact with the road?
42. You're stuck in the exit row on a long flight, and you suddenly worry that your seatmate, who's next to the window, might pull the emergency window inward while you're in flight. The window measures 50 cm by 60 cm. Cabin pressure is 0.8 atm, and atmospheric pressure at the plane's altitude is 0.2 atm. Should you worry?
43. A vertical tube 1.7 cm in diameter and open at the top contains 5.6 g of oil (density 0.82 g/cm^3) floating on 5.6 g of water. Find the gauge pressure (a) at the oil–water interface and (b) at the bottom.
44. Dam breaks present a serious risk of widespread property damage and loss of life. You're asked to assess a 1500-m-wide dam holding back a lake 95 m deep. The dam was built to withstand a force of 100 GN, which is supposed to be at least 50% over the force it actually experiences. Should the dam be reinforced? (*Hint*: You'll need your calculus skills.)
45. A U-shaped tube open at both ends contains water and a quantity of oil occupying a 2.0-cm length of the tube, as shown in Fig. 15.24. If the oil's density is 92% of water's, what's the height difference h ?

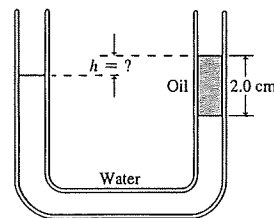


FIGURE 15.24 Problem 45

46. You're a robotics engineer designing a hydraulic system to move a robotic arm. The hydraulic cylinder that drives the arm has diameter 6.0 cm and can exert a maximum force of 6.1 kN. Hydraulic tubing comes rated in multiples of $1/2 \text{ MPa}$, and, for safety, you're to specify tubing capable of withstanding 50% greater pressure than it will ever experience in use. What pressure rating do you specify?
47. Hydraulic brake systems are increasingly common on bicycles, especially those using disc brakes. The Magura MT4 brake system, widely used on mountain bikes, has a small piston and cylinder of diameter 1.04 cm connected to the brake actuation lever on the handlebars, and a larger 2.10-cm-diameter piston/cylinder that pushes the brake pads against the rotating brake disc. (a) What force must be applied to the smaller piston to produce a force of 3.25 kN on the larger piston? (b) If the smaller piston moves a maximum of 8.80 mm, what's the corresponding motion of the larger piston? (c) How much work is done on each piston as they undergo the motions of part (b)?
48. Archimedes purportedly used his principle to verify that the king's crown was pure gold by weighing the crown submerged in water. Suppose the crown's actual weight was 25.0 N. What would be its apparent weight if it were made of (a) pure gold and (b) 75% gold and 25% silver, by volume? The densities of gold, silver, and water are 19.3 g/cm^3 , 10.5 g/cm^3 , and 1.00 g/cm^3 , respectively.
49. You're testifying in a drunk-driving case for which a blood alcohol measurement is unavailable. The accused weighs 140 lb, and would be legally impaired after consuming 36 oz of beer. The accused was observed at a beach party where a keg with interior diameter 40 cm was floating in the lake to keep it cool. After the accused's drinking stint, the keg floated 1.2 cm higher than before. Beer's density is essentially that of water. Does your testimony help or hurt the accused's case?
50. A glass beaker measures 12 cm high by 7.0 cm in diameter. Empty, it floats in water with one-third of its height submerged. How many 16-g rocks can be placed in the beaker before it sinks?
51. A typical supertanker has mass $2.0 \times 10^6 \text{ kg}$ and carries twice that much oil. If 8.0 m of the ship is submerged when it's empty, what's the minimum water depth needed for it to navigate when full? Assume the sides of the ship are vertical.
52. A balloon contains gas of density ρ_g and is to lift a mass M , including the balloon but not the gas. Show that the minimum mass of gas required is $m_g = M\rho_g/(\rho_a - \rho_g)$, where ρ_a is the atmospheric density.
53. (a) How much helium (density 0.18 kg/m^3) is needed to lift a balloon carrying two people, if the total mass of people, basket, and balloon

- (but not gas) is 280 kg? (b) Repeat for a hot-air balloon whose air density is 10% less than that of the surrounding atmosphere.
54. A newfangled swim float, the MaxRaft by WaterMat, measures $1.8\text{ m} \times 2.4\text{ m}$ and is only 10 cm thick. Its mass is 20 kg. How many 50-kg children can the float accommodate before the water comes over its top surface? Assume the float stays perfectly level.
55. If the blood pressure in the unobstructed artery of Exercise 31 is 16 kPa gauge (about 120 mm of mercury, the unit commonly reported by doctors), what will it be at the clot? (Note: Blood's density is 1.06 g/cm^3 .)
56. You're a consultant for maple syrup producers. They tap maple trees and collect sap with plastic tubing that connects to a common pipe delivering sap to an evaporator. There it's boiled to produce thick, tasty syrup. The system can be modeled as a pipe with one end, of cross-sectional area A , exposed to atmospheric pressure. The pipe drops through a vertical distance h_1 while its area decreases to $A/2$, as shown in Fig. 15.25. A small vertical glass tube extends from the lower portion, as shown, and is open to atmospheric pressure. You're asked to provide a formula for the volume flow rate of the sap as a function of the height h_2 of sap in the tube.

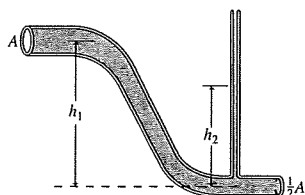


FIGURE 15.25 Problem 56

57. The water in a garden hose is at 120-kPa gauge pressure and is moving at negligible speed. The hose terminates in a sprinkler consisting of many small holes. Find the maximum height reached by the water emerging from the holes.
58. The venturi flowmeter shown in Fig. 15.26 is used to measure the flow rate of water in a solar collector system. The flowmeter is inserted in a pipe with diameter 1.9 cm; at the venturi, the diameter is 0.64 cm. The manometer tube contains oil with density 0.82 times that of water. If the difference in oil levels on the two sides of the manometer tube is 1.9 cm, what's the volume flow rate?

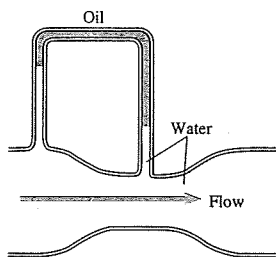


FIGURE 15.26 Problem 58

59. A 1.3-cm-diameter venturi flowmeter is inserted in a 2.3-cm-diameter pipe carrying water (density 1000 kg/m^3). Find (a) the flow speed in the pipe and (b) the volume flow rate if the pressure difference between venturi and unconstricted pipe is 15 kPa.
60. A balloon's mass is 1.6 g when it's empty. It's inflated with helium (density 0.18 kg/m^3) to form a sphere 28 cm in diameter. How many 0.63-g paper clips can you hang from the balloon before it loses buoyancy?
61. Blood with density 1.06 g/cm^3 and 10-kPa gauge pressure flows through an artery at 31 cm/s. It encounters a plaque deposit where the pressure drops by 6%. What fraction of the artery's area is obstructed?
62. A venturi flowmeter in an oil pipeline has a radius half that of the pipe. Oil flows in the unconstricted pipe at 1.9 m/s. If the pressure difference between unconstricted flow and venturi is 16 kPa, what's the oil's density?
63. Homes in rural areas of the developed world generally get their water from wells. Two types of pumps are used to move water from the well to the house. For shallow wells, a pump mounted inside the home produces a partial vacuum that results in air pressure pushing water out of the well. Deeper wells, on the other hand, require submerged pumps, because there's a limit to

how high air pressure can lift the well water. Suppose a given shallow-well pump can produce a partial vacuum whose pressure is 33.3% of standard atmospheric pressure. What's the maximum well depth at which this pump will function?

64. In 2012, film producer James Cameron (*Terminator*, *Titanic*, *Avatar*) rode his submersible *Deepsea Challenger* to the bottom of the 11-km-deep Marianas Trench, the deepest spot in Earth's oceans. Cameron could barely fit into *Deepsea Challenger*'s crew compartment, a steel sphere with inside diameter 109 cm and walls 6.4 cm thick. Find the total pressure force exerted on the sphere at the bottom of the trench. (The total force is the sum of all pressure forces without regard to direction; it's not the same as the buoyancy force, which is the *net* pressure force—a vectorial sum.)
65. A probe descending through Mars' atmosphere records pressure as a function of altitude; the data are in the table below. Plot the natural logarithm of the pressure versus altitude and fit a line to your plotted points. Mars' atmospheric pressure is governed by the same equation that describes Earth's; see Problem 72. Use your fitted line, in connection with that equation, to determine (a) Mars' surface pressure and (b) the scale height h_0 .

Altitude (km)	10	20	30	40	50	60
Pressure (Pa)	242	98.7	37.6	16.2	7.21	2.38

66. Water emerges from a faucet of diameter d_0 in steady, near-vertical flow with speed v_0 . Show that the diameter of the falling water column is given by $d = d_0 [v_0^2 / (v_0^2 + 2gh)]^{1/4}$, where h is the distance below the faucet (Fig. 15.27).

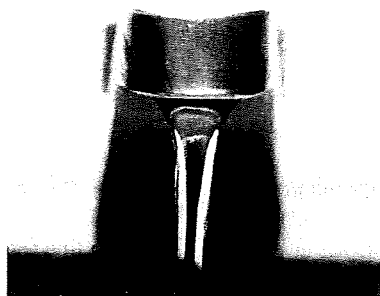


FIGURE 15.27 Problem 66

67. Assuming normal atmospheric pressure, how massive an object can a 4.0-cm-diameter suction cup support on a vertical wall if the coefficient of friction between cup and wall is 0.68?
68. Figure 15.28 shows a simplified diagram of a Pitot tube, used for measuring aircraft speeds. The tube is mounted on the aircraft with opening A at right angles to the flow and opening B pointing into the flow. The gauge prevents airflow through the tube. Use Bernoulli's equation to show that the plane's speed relative to the air is $v = \sqrt{2 \Delta p / \rho}$, where Δp is the pressure difference between the tubes and ρ is the density of air. (Hint: The flow must be stopped at B , but continues past A with its normal speed.)

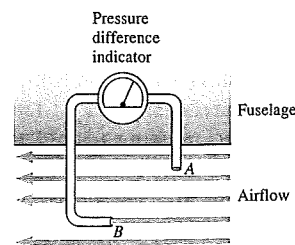


FIGURE 15.28 Problem 68

69. At a hearing on a proposed wind farm, a wind-energy advocate says an installation of 900 turbines, with blade diameter 110 m, could displace a 1-GW nuclear power plant. You're asked if that's really possible. How do you answer, given an average wind speed of 10 m/s and a turbine power output that averages 30% of the theoretical maximum?

70. A pencil is weighted so it floats vertically with length L submerged. It's pushed vertically downward without being totally submerged, then released. Show that it undergoes simple harmonic motion with period $T = 2\pi\sqrt{L/g}$.

71. A can of height h and cross-sectional area A_0 is initially full of water. A small hole of area $A_1 \ll A_0$ is cut in the bottom of the can. Find an expression for the time it takes all the water to drain from the can. (Hint: Call the water depth y , use the continuity equation, and integrate.)

72. In the approximation of constant atmospheric temperature, density and pressure in Earth's atmosphere are related by $\rho = p/h_0g$, where $h_0 = 8.2$ km is a constant called the *scale height* and g is the gravitational acceleration. (a) Integrate Equation 15.2 for this case to show that atmospheric pressure as a function of height h above the surface is given by $p = p_0 e^{-h/h_0}$, where p_0 is the surface pressure. (b) At what height will the pressure have dropped to half its surface value?

73. (a) Use the result of Problem 72 to express Earth's atmospheric density as a function of height. (b) Use your result from (a) to find the height below which half of Earth's atmospheric mass lies (this will require integration).

74. A circular pan of liquid with density ρ is centered on a horizontal turntable rotating with angular speed ω , as shown in Fig. 15.29. Atmospheric pressure is p_a . Find expressions for (a) the pressure at the bottom of the pan and (b) the height of the liquid surface, both as functions of the distance r from the axis, given that the height at the center is h_0 .

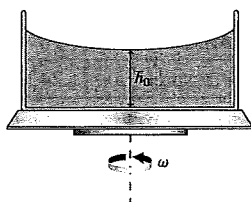


FIGURE 15.29 Problem 74

75. Find the torque that the water exerts about the bottom edge of the dam in Problem 44.
76. One vertical wall of a swimming pool is a regular trapezoid, with its bottom 15 m long and its top 22 m long. The pool is 3.3 m deep, and it's full to the brim with water. Find the pressure force the water exerts on this side of the pool.
77. You're a private investigator assisting a fine food manufacturer in tracking down counterfeit deluxe salad dressing. The genuine dressing is, by volume, one part traditional Reggio Emilia balsamic vinegar (density 1.20 g/cm^3) to three parts olive oil (density 0.92 g/cm^3). The counterfeit dressing uses ordinary balsamic vinegar (density 1.06 g/cm^3). You measure the density of a dressing sample and find it to be 0.99 g/cm^3 . Has the dressing been altered?
78. A plumber comes to your ancient apartment building where you have a part-time job as caretaker. He's checking the hot-water heating system, and notes that the water pressure in the basement is 18 psi. He asks, "How high is the building?" "Three stories, each about 11 feet," you reply. "OK, about 33 feet," he says, pausing to do some calculations in his head. "The pressure is fine," he declares. On what basis did he come to that conclusion?

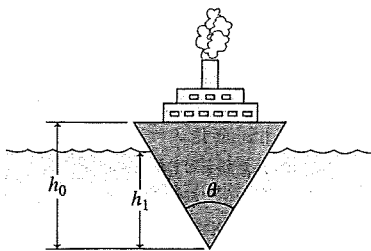


FIGURE 15.30 Problem 79

79. Your class in naval architecture is working on the design for a ship with a V-shaped cross section, as shown in Fig. 15.30. The ship

has total length L and keel-to-deck height h_0 . When empty, the distance from waterline to keel is h_1 . You're asked for the maximum load the ship can carry below deck if water is not to come over the deck. Answer in terms of h_0 , h_1 , L , θ , and the water density ρ .

Passage Problems

Arterial stenosis is a constriction of an artery, often due to plaque buildup on the artery's inner walls. Serious medical conditions can result, depending on the affected artery. Stenosis of the carotid arteries that supply blood to the brain is a leading cause of stroke, while stenosis of the renal arteries can lead to kidney failure. Pulmonary artery stenosis results from birth defects, and can result in insufficient oxygen supply. Because the heart has to work harder to get blood through a constricted artery, stenosis can contribute to high blood pressure.

In answering the questions below, assume steady flow (which is true in arteries only on short timescales).

80. How does the volume flow rate of blood at a stenosis compare with the rate in the surrounding artery?
- lower
 - the same
 - higher
81. How does the blood flow speed at a stenosis compare with the speed in the surrounding artery?
- lower
 - the same
 - higher
82. Which of the following medical problems is more likely to occur?
- An artery might collapse because of lower blood pressure at the stenosis.
 - An artery might burst because of higher blood pressure at the stenosis.
 - Neither; pressure at the stenosis is the same as in the surrounding artery.
83. If the artery has circular cross section even at the stenosis, but the diameter at the stenosis is half that in the surrounding artery, the blood flow speed in the stenosis will be
- one-fourth that in the surrounding artery.
 - one-half that in the surrounding artery.
 - the same as in the surrounding artery.
 - $\sqrt{2}$ times that in the surrounding artery.
 - four times that in the surrounding artery.

Answers to Chapter Questions

Answer to Chapter Opening Question

Because the density of ice is only slightly less than that of water.

Answers to GOT IT? Questions

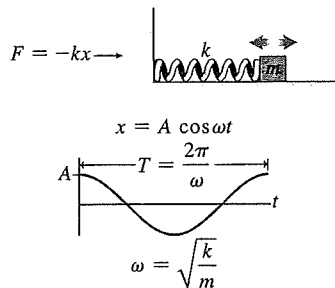
- 15.1 (c)
- 15.2 (c) $\vec{F}$ moves the small piston a lot farther than the upward pressure force moves the large piston; the products of force and displacement are the same for both pistons, so the work done is the same.
- 15.3 (a)
- 15.4 (a) over the top where the streamlines are closer together
- 15.5 $h_1 > h_4 > h_2 > h_3$ reflecting higher pressure with lower flow speed

Summary

Part Two has extended Newtonian mechanics to systems that undergo oscillatory motion and wave motion or that involve the motion of

Oscillatory motion describes the back-and-forth motion of a system disturbed from a stable equilibrium.

When the force or torque tending to restore equilibrium is directly proportional to the displacement, the result is simple harmonic motion.

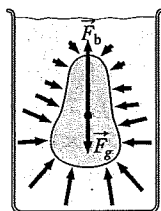


When waves overlap, the result is **interference**, which is constructive when the waves reinforce and destructive when they tend to cancel.



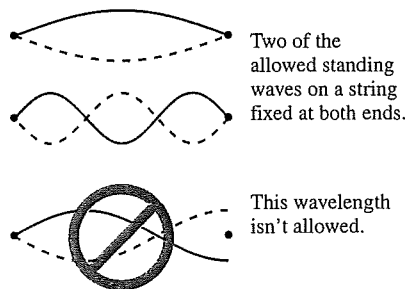
Nodal lines: destructive interference
Large amplitude: constructive interference

Fluids in **hydrostatic equilibrium** exhibit a depth-dependent pressure that results in an upward buoyancy force $\vec{F}_b$.



Archimedes' principle states that the buoyancy force is equal to the weight of the displaced fluid.

Standing waves occur when the medium has limited extent. Only certain wavelengths and frequencies are allowed, depending on the medium's length:



Two of the allowed standing waves on a string fixed at both ends.

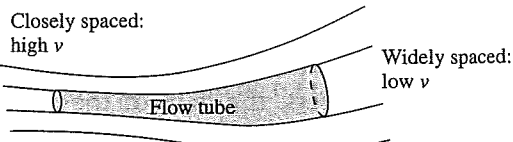
This wavelength isn't allowed.

Moving fluids obey conservation of mass and, in the absence of fluid friction (viscosity), they also conserve energy.

In **fluid dynamics**, the continuity equation and Bernoulli's equation express these conservation laws. Both equations hold along a flow tube:

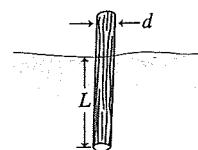
Continuity: $\rho v A = \text{constant}$
Bernoulli:

$$p + \frac{1}{2} \rho v^2 + \rho g y = \text{constant}$$



Part Two Challenge Problem

A cylindrical log of total mass M and uniform diameter d has an uneven mass distribution that causes it to float in a vertical position, as shown in the figure. (a) Find an expression for the length L of the submerged portion of the log when it's floating in equilibrium, in terms of M , d , and the water density ρ . (b) If the log is displaced vertically from its equilibrium position and released, it will undergo simple harmonic motion. Find an expression for the period of this motion, neglecting viscosity and other frictional effects.



Oscillations, Waves, and Fluids

fluids. Behind these more complex motions are the fundamental concepts of force, mass, and energy and their roles in characterizing motion.

A **wave** is a propagating disturbance that carries energy but not matter.

Simple harmonic waves are sinusoidal:

$$y(x, t) = A \cos(kx - \omega t)$$

Angular frequency: $\omega = 2\pi f$

Wave number: $k = \frac{2\pi}{\lambda}$

Wave period: $T = \frac{1}{f}$

Wave speed: $v = \frac{\omega}{k} = \frac{\lambda}{T} = f\lambda$

